Serjan Kaur

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Education

Western University — London, ON
Bachelor of Science in Computer Science, Co-op — *2023 – 2027*

- Relevant Coursework: Software Engineering, Database Systems, Data Structures & Algorithms, Object-Oriented Programming

Experience

Data Engineering Intern — AI/ML + EDGE — Sep. 2026 – Dec. 2026
CIBC — *Toronto, ON*

- Deploy **LLM-powered applications and model-serving endpoints** on Microsoft Azure and Databricks, giving internal teams secure, scalable access to AI models in a regulated banking environment.
- Build and configure **REST APIs through Azure API Management (APIM)** that connect multiple LLMs to internal applications and enterprise workflows.
- Test and troubleshoot LLM deployments, and create **reusable integration workflows** with AI/ML teams so new models follow a repeatable, governed process.

AI Automation Engineering Intern — Jul. 2026 – Sep. 2026
Mason Technology Corporation — *Remote*

- Built **agentic AI workflows with Claude**, Python, and REST APIs that call tools, retrieve business data, and run multi-step tasks like prospect research and lead enrichment.
- Worked directly with the **founding team** to map manual processes, define what each automation should do, and turn requirements into working prototypes.
- Tested agents against real business cases and **iterated on prompts, tool calls, and workflow logic** based on team feedback to improve reliability and reduce manual work.

Technical Intern — AI Systems Development — Mar. 2026 – Jun. 2026
Equintel — *Remote*

- Developed **Python NLP and data-processing pipelines** that turn unstructured sustainability reports and regulatory disclosures into structured data for AI-driven ESG assessments.
- Contributed to **AI retrieval workflows** for natural-language exploration of regulatory data, and added **validation and QA checks** on AI outputs to make assessments more trustworthy.

Software Developer & AI Automation Lead — Jan. 2024 – Jan. 2026
Pioneer Medcare — *Toronto, ON*

- Partnered with business stakeholders to find operational bottlenecks, then **prototyped, tested, and refined automation workflows** through repeated rounds of user feedback.
- Built **AI-assisted delivery route optimization**, improving **driver efficiency by 60%** by cutting manual coordination and route-planning delays.
- Built **AI-powered ordering and workflow automation** for B2B customers, cutting **order processing time by 70%**.
- Designed **REST APIs and backend services** (Node.js, SQL, AWS) linking inventory and orders with Amazon and eBay.

Projects

Canadian Housing Burden Prediction | *Python, XGBoost, Scikit-Learn, K-Means, SHAP* — GitHub

- Co-built (team of 3) an **end-to-end ML pipeline** on proprietary Environics Analytics data covering **56,857 dissemination areas**, delivered as a consultancy report for decision-makers.
- Identified and removed **56 leakage-prone features** that would have inflated accuracy, fitting all preprocessing on training data only so results reflect real-world performance.
- Compared Elastic Net and XGBoost (**0.58 test** R^2), used **SHAP** to explain drivers and K-Means to find a high-burden segment, and reported the model's limits openly.

Constraint-Based Exam Scheduling Engine | *Python, OR-Tools CP-SAT, Streamlit* — Live Demo

- Built a **CP-SAT optimization engine** that generates conflict-free exam schedules for about **1,000 students and 100 courses**, turning a fairness judgment into rules that cap daily exam load.
- Shipped a **Streamlit app** so non-technical users can generate timetables, explore schedules, and review conflict reports.

Technical Skills

AI & Agents: Claude, LLMs, AI Agents, Agentic Workflows, Tool Calling, RAG, NLP, Prompt Engineering
Cloud & Deployment: Microsoft Azure, Azure APIM, Databricks, AWS, Docker, Model Serving, CI/CD
Languages & Frameworks: Python, SQL, JavaScript, TypeScript, Java, C++, Node.js, Flask, Streamlit
APIs, Data & ML: REST APIs, ETL, PostgreSQL, MySQL, Pandas, Scikit-Learn, XGBoost, OR-Tools, Git, Postman

Leadership

Vice President, Events — May 2025 – Apr. 2026
Western Cyber Society — *London, ON*

- Planned and ran the club's event lineup, including **IBM Night**, a **resume workshop**, and a large member social, connecting students with industry professionals and each other.
- Helped organize the **Canadian Tech Summit** (**200+ attendees, 40 project teams**) showcasing student-built tech.